Linear Algebra Foundations for Systems

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Unit 14: Vectors, Transformations, Determinants, and Eigentheory

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Unit 14 at a Glance: Linear Algebra for Systems

Before coupled differential equations can be solved, the language of linear algebra has to be in place. A matrix is read geometrically: its columns say where the basis vectors land, its determinant says how it scales area and whether it flips orientation, and its eigenvectors mark the special directions it merely stretches. The payoff is the eigenvalue method — guessing $\mathbf{x} = e^{\lambda t}\mathbf{v}$ turns the system $\mathbf{x}' = A\mathbf{x}$ into the algebraic eigenvalue problem $A\mathbf{v} = \lambda\mathbf{v}$.

- **Vectors and transformations** — linear combinations, span, basis, and the matrix whose *columns are the images of the basis vectors.*
- **The determinant** — the signed area-scaling factor $ad - bc$; zero means collapse, negative means an orientation flip.
- **Change of basis** — the similarity transform $P^{-1}AP$ rewrites the same map in new coordinates; an eigenbasis makes it diagonal.
- **Eigentheory** — $A\mathbf{v} = \lambda\mathbf{v}$, the characteristic equation $\det(A - \lambda I) = 0$, and the trace/determinant shortcut for 2×2.
- **Eigenvalues through ODEs** — $\mathbf{x} = e^{\lambda t}\mathbf{v}$ decouples $\mathbf{x}' = A\mathbf{x}$ into scalar modes, the gateway to Unit 15.

14.1 Vectors, Span, and Linear Transformations

Linear combinations, span, and basis; linear transformations and the matrices that encode them; columns as images of the basis vectors; composition as matrix product.

Linear Combination, Span, and Basis

$$a\mathbf{v} + b\mathbf{w} \quad (a, b \text{ scalars}); \quad \operatorname{span}\{\mathbf{v}_i\} = \left\{ \textstyle\sum_i c_i\mathbf{v}_i \right\}; \quad \text{basis} \iff \text{independent } \textbf{and} \text{ spanning.}$$

Linearity Condition

$$T(a\mathbf{u} + b\mathbf{v}) = a\,T(\mathbf{u}) + b\,T(\mathbf{v}) \quad \text{for all scalars } a, b \text{ and vectors } \mathbf{u}, \mathbf{v}.$$

The Matrix of a Transformation

$$A = \big[\, T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \,\big], \qquad T(\mathbf{x}) = A\mathbf{x}, \qquad (T_2 \circ T_1) \leftrightarrow A_2 A_1.$$

Method — Build the Matrix of a Linear Map

1. Apply T to each standard basis vector $\mathbf{e}_j$.
2. Record each image $T(\mathbf{e}_j)$ as the j-th *column*.
3. Assemble the columns into the matrix A.
4. Read off the action as $T(\mathbf{x}) = A\mathbf{x}$ and verify on a test vector.

Worked Example — Apply a Matrix to a Vector

Let A have rows $(1, 2)$ and $(3, 4)$. Compute the image of $\mathbf{x} = (1, 1)$ by dotting each row with $\mathbf{x}$.

$$A\mathbf{x} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\text{row 1: } (1, 2) \cdot (1, 1) = 1 + 2 = 3$$

$$\text{row 2: } (3, 4) \cdot (1, 1) = 3 + 4 = 7$$

$$A\mathbf{x} = (3,\ 7)$$

Key Takeaway

A linear transformation is completely determined by where it sends the basis. The columns of A are exactly the images $T(\mathbf{e}_j)$, and composing two maps corresponds to *multiplying* their matrices (right-to-left).

Common Pitfall

A map with a nonzero shift, $T(\mathbf{u}) = \mathbf{u} + \mathbf{c}$ with $\mathbf{c} \neq \mathbf{0}$, is affine, not linear — it moves the origin. Linearity *requires* $T(\mathbf{0}) = \mathbf{0}$. Also, matrix multiplication is not commutative: $A_2A_1 \neq A_1A_2$ in general, so composition order matters.

14.2 The Determinant and Change of Basis

The determinant as a signed area-scaling factor; the formula $ad - bc$; what zero and negative determinants mean; change-of-basis matrices and the similarity transform $P^{-1}AP$.

The 2×2 Determinant

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc \;=\; \text{signed factor by which the map scales area.}$$

Reading the Determinant Geometrically

$$|\det A| = \text{area scaling}, \quad \det A = 0 \Rightarrow \text{collapse (dependent columns)}, \quad \det A < 0 \Rightarrow \text{orientation reversed.}$$

Change of Basis and Similarity

$$B = P^{-1}AP, \qquad \text{columns of } P = \text{new basis vectors in the old coordinates}; \quad \text{eigenbasis} \Rightarrow B \text{ diagonal.}$$

Method — Re-express a Map in a New Basis

1. Place the new basis vectors (written in old coordinates) as the columns of P.
2. Compute the inverse P^{-1}.
3. Form the similarity transform $B = P^{-1}AP$.
4. If the columns of P are eigenvectors of A, then B is diagonal with the eigenvalues on the diagonal.

Worked Example — Determinant and Its Meaning

Compute the determinant of A with rows $(3, 1)$ and $(2, 4)$, then interpret it geometrically.

$$\det A = ad - bc = (3)(4) - (1)(2)$$

$$\det A = 12 - 2 = 10$$

$$|\det A| = 10 \Rightarrow \text{areas are scaled by a factor of } 10$$

$$\det A > 0 \Rightarrow \text{orientation is preserved (no flip).}$$

Key Takeaway

The determinant packages two facts in one number: its magnitude is the area-scaling factor and its sign records orientation. Because a change of basis only relabels coordinates, *similar* matrices $P^{-1}AP$ share the same determinant, trace, and eigenvalues.

Common Pitfall

Do not confuse a negative determinant with a zero determinant. Negative still means invertible — the plane is merely flipped. Only $\det A = 0$ collapses a dimension and makes A singular (columns linearly dependent).

14.3 Eigenvalues and Eigenvectors

The eigenvalue equation $A\mathbf{v} = \lambda\mathbf{v}$; the characteristic equation $\det(A - \lambda I) = 0$; trace as the sum and determinant as the product of eigenvalues; the trace/determinant shortcut for 2×2 matrices.

The Eigenvalue Equation

$$A\mathbf{v} = \lambda\mathbf{v}, \quad \mathbf{v} \neq \mathbf{0}: \quad A \text{ only } \textit{stretches } \mathbf{v} \text{ along its own line, by the factor } \lambda.$$

The Characteristic Equation

$$\det(A - \lambda I) = 0; \qquad 2 \times 2: \quad \lambda^2 - (\operatorname{tr} A)\,\lambda + \det A = 0.$$

Trace–Determinant Shortcut

$$\lambda = m \pm \sqrt{m^2 - p}, \quad m = \tfrac{1}{2}\operatorname{tr} A, \quad p = \det A; \qquad \operatorname{tr} A = \lambda_1 + \lambda_2, \quad \det A = \lambda_1\lambda_2.$$

Method — Find Eigenvalues and Eigenvectors

1. Form $A - \lambda I$.
2. Set $\det(A - \lambda I) = 0$ and solve for λ (for 2×2, use $\lambda^2 - (\operatorname{tr} A)\lambda + \det A = 0$).
3. For each λ, solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$ for the eigenvector $\mathbf{v}$.
4. Verify $A\mathbf{v} = \lambda\mathbf{v}$.

Worked Example — Eigenvalues and Eigenvectors of a Symmetric Matrix

For A with rows $(2, 1)$ and $(1, 2)$, use $\operatorname{tr} A = 4$ and $\det A = 3$, then find an eigenvector for each eigenvalue.

$$\lambda = m \pm \sqrt{m^2 - p} = 2 \pm \sqrt{4 - 3} = 2 \pm 1 \;\Rightarrow\; \lambda_1 = 3,\; \lambda_2 = 1$$

$$\lambda_1 = 3: \;(A - 3I)\mathbf{v} = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}\mathbf{v} = \mathbf{0} \;\Rightarrow\; \mathbf{v}_1 = (1, 1)$$

$$\lambda_2 = 1: \;(A - I)\mathbf{v} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\mathbf{v} = \mathbf{0} \;\Rightarrow\; \mathbf{v}_2 = (1, -1)$$

$$\text{Check: } A(1, 1) = (3, 3) = 3(1, 1) \checkmark$$

Key Takeaway

Eigenvectors are the special directions a matrix leaves invariant — it only scales them, by the eigenvalue. In the eigenbasis the matrix becomes diagonal, which is exactly what makes powers, exponentials, and differential equations easy.

Common Pitfall

The characteristic equation is $\det(A-\lambda I)=0$, not $\det A-\lambda=0$, and the 2×2 form carries a *minus* sign on the trace: $\lambda^2-(\operatorname{tr} A)\lambda+\det A$. Handy check: for diagonal or triangular matrices the eigenvalues sit on the diagonal — the off-diagonal entries do not change them.

14.4 Eigenvalues Through Differential Equations

The trial solution $\mathbf{x}=e^{\lambda t}\mathbf{v}$ that reduces $\mathbf{x}'=A\mathbf{x}$ to the eigenvalue problem; one exponential mode per eigenpair; the general solution by superposition; growth, decay, and oscillation from the sign and type of the eigenvalues.

The Eigen-Ansatz

$$\text{Try } \mathbf{x}=e^{\lambda t}\mathbf{v}: \quad \mathbf{x}'=\lambda e^{\lambda t}\mathbf{v}=Ae^{\lambda t}\mathbf{v} \;\Rightarrow\; A\mathbf{v}=\lambda\mathbf{v}.$$

General Solution (Distinct Eigenvalues)

$$\mathbf{x}(t)=c_1e^{\lambda_1 t}\mathbf{v}_1+c_2e^{\lambda_2 t}\mathbf{v}_2 \quad \text{(superposition of the eigen-modes).}$$

Stability from the Eigenvalues

$$\lambda<0:\ \text{decay},\quad \lambda>0:\ \text{growth},\quad \lambda=\alpha\pm i\beta:\ \text{oscillation (decays if } \alpha<0).$$

Method — Solve $\mathbf{x}'=A\mathbf{x}$ by Eigenvalues

1. Find the eigenvalues of A from $\det(A-\lambda I)=0$.
2. Find an eigenvector $\mathbf{v}_i$ for each eigenvalue λ_i.
3. Form the eigen-modes $e^{\lambda_i t}\mathbf{v}_i$.
4. Superpose: $\mathbf{x}=\sum_i c_ie^{\lambda_i t}\mathbf{v}_i$, then fix the c_i from the initial conditions.

Worked Example — Solve a Coupled System

Solve $\mathbf{x}'=A\mathbf{x}$ for A with rows $(2,1)$ and $(1,2)$, reusing the eigenpairs $\lambda_1=3,\ \mathbf{v}_1=(1,1)$ and $\lambda_2=1,\ \mathbf{v}_2=(1,-1)$ from 14.3.

Each eigenpair gives a mode: $e^{3t}(1,1)$ and $e^{t}(1,-1)$

$$\mathbf{x}(t)=c_1e^{3t}\begin{pmatrix}1\\1\end{pmatrix}+c_2e^{t}\begin{pmatrix}1\\-1\end{pmatrix}$$

Both $\lambda>0 \Rightarrow$ every trajectory grows; the origin is an unstable node.

Key Takeaway

The eigenvalue method decouples a coupled linear system into independent scalar modes. Along each eigenvector A acts as multiplication by λ, so $\mathbf{x}' = A\mathbf{x}$ collapses to $x' = \lambda x$ with solution $e^{\lambda t}$ — the foundation of every technique in Unit 15.

Common Pitfall

In the superposition each exponential must ride its own eigenvector: $c_1 e^{\lambda_1 t}\mathbf{v}_1 + c_2 e^{\lambda_2 t}\mathbf{v}_2$, never a crossed pairing. And long-term behavior is set by the sign of $\mathrm{Re}(\lambda)$ — a single positive eigenvalue makes the whole system unstable.

Practice Problems

Work the following ten problems in order. They climb through linear combinations and bases, the matrix of a transformation, the determinant and its geometric meaning, the similarity transform, the characteristic equation and the trace/determinant shortcut, eigenvectors, the eigen-ansatz, and a capstone that solves a coupled system $\mathbf{x}' = A\mathbf{x}$ end to end. Complete solutions for all problems follow.

1. **(Linear combination)** Compute the linear combination $2(1, 0) + 3(0, 1)$.

2. **(Basis test)** State the two conditions a set must meet to be a basis, then decide whether $(1, 0)$ and $(2, 0)$ form a basis of $\mathbb{R}^2$.

3. **(Matrix of a transformation)** For the matrix A with rows $(1, 2)$ and $(3, 4)$, compute the image $A\mathbf{x}$ of $\mathbf{x} = (1, 1)$.

4. **(Determinant, compute)** Find the determinant of the matrix with rows $(3, 1)$ and $(2, 4)$, and state the factor by which it scales area.

5. **(Determinant, meaning)** Compute the determinant of the matrix with rows $(1, 2)$ and $(2, 4)$. Is the matrix invertible? What does this say about its columns?

6. **(Change of basis)** Write the formula that expresses a matrix A in the basis given by a change-of-basis matrix P, and state what makes the result diagonal.

7. **(Characteristic equation)** Write the 2×2 characteristic equation in terms of the trace and determinant, then find the eigenvalues of the matrix with rows $(2, 1)$ and $(1, 2)$.

8. **(Eigenvectors)** For that same matrix (rows $(2, 1)$, $(1, 2)$), find an eigenvector for each eigenvalue.

9. **(Eigen-ansatz)** Show that substituting $\mathbf{x} = e^{\lambda t}\mathbf{v}$ into $\mathbf{x}' = A\mathbf{x}$ produces the eigenvalue equation $A\mathbf{v} = \lambda\mathbf{v}$.

10. **(Capstone — solve a system)** Using the eigenpairs from Problems 7–8, write the general solution of $\mathbf{x}' = A\mathbf{x}$ for A with rows $(2, 1)$ and $(1, 2)$, and classify the equilibrium at the origin.

Complete Solutions

Solution 1. Scale each vector and add componentwise:

$$2(1,0)+3(0,1)=(2,0)+(0,3)=(2,3).$$

The scalars stay attached to their own directions; they do not merge into a single axis.

Solution 2. A basis must be *linearly independent* and must *span* the space. The vectors $(1,0)$ and $(2,0)$ are parallel, since $(2,0)=2(1,0)$, so they are dependent and span only the x-axis:

$$\operatorname{span}\{(1,0),(2,0)\}=\{(t,0)\}\neq\mathbb{R}^2.$$

They are **not** a basis of $\mathbb{R}^2$.

Solution 3. Dot each row with the input $\mathbf{x}=(1,1)$:

$$A\mathbf{x}=\begin{pmatrix}1&2\\3&4\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix}=\begin{pmatrix}1+2\\3+4\end{pmatrix}=\begin{pmatrix}3\\7\end{pmatrix}.$$

Solution 4. Use $\det = ad-bc$ with $a=3, b=1, c=2, d=4$:

$$\det A=(3)(4)-(1)(2)=12-2=10.$$

The map scales every area by a factor of $|\det A|=10$, and since $\det A>0$ it preserves orientation.

Solution 5. Compute the determinant:

$$\det\begin{pmatrix}1&2\\2&4\end{pmatrix}=(1)(4)-(2)(2)=4-4=0.$$

A zero determinant means the matrix is **not invertible**: it collapses the plane onto a line, so its columns are linearly dependent (indeed $(2,4)=2(1,2)$).

Solution 6. The similarity transform rewrites A in the new coordinates:

$$B=P^{-1}AP,$$

where the columns of P are the new basis vectors in the old coordinates. If those columns are eigenvectors of A, then B is **diagonal**, with the eigenvalues on the diagonal.

Solution 7. The characteristic equation is

$$\lambda^2-(\operatorname{tr}A)\lambda+\det A=0.$$

For rows $(2,1),(1,2)$: $\operatorname{tr}A=4$, $\det A=3$, so

$$\lambda^2-4\lambda+3=(\lambda-3)(\lambda-1)=0 \;\Rightarrow\; \lambda=3,\;1.$$

Solution 8. Solve $(A-\lambda I)\mathbf{v}=\mathbf{0}$ for each eigenvalue:

$$\lambda=3:\;\begin{pmatrix}-1&1\\1&-1\end{pmatrix}\mathbf{v}=\mathbf{0}\Rightarrow\mathbf{v}_1=(1,1);$$

$$\lambda = 1: \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \mathbf{v} = \mathbf{0} \Rightarrow \mathbf{v}_2 = (1, -1).$$

Solution 9. Differentiate the trial $\mathbf{x} = e^{\lambda t}\mathbf{v}$ (with $\mathbf{v}$ constant) and substitute:

$$\mathbf{x}' = \lambda e^{\lambda t}\mathbf{v}, \qquad A\mathbf{x} = e^{\lambda t} A\mathbf{v}.$$

Setting $\mathbf{x}' = A\mathbf{x}$ and cancelling the nonzero scalar $e^{\lambda t}$ leaves

$$\lambda \mathbf{v} = A\mathbf{v},$$

exactly the eigenvalue problem.

Solution 10. Each eigenpair contributes one exponential mode; superpose them:

$$\mathbf{x}(t) = c_1 e^{3t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Both eigenvalues are real and positive, so every trajectory grows without bound: the origin is an **unstable node** (a source).

Unit Key Takeaway

Every problem above is one geometric story: a matrix *moves* space. Its columns say where the basis lands, its determinant says how area is scaled and whether orientation flips, and its eigenvectors mark the directions it merely stretches. Feeding those eigenpairs into $\mathbf{x} = e^{\lambda t}\mathbf{v}$ turns a coupled system into independent exponential modes — the exact machinery Unit 15 runs on.